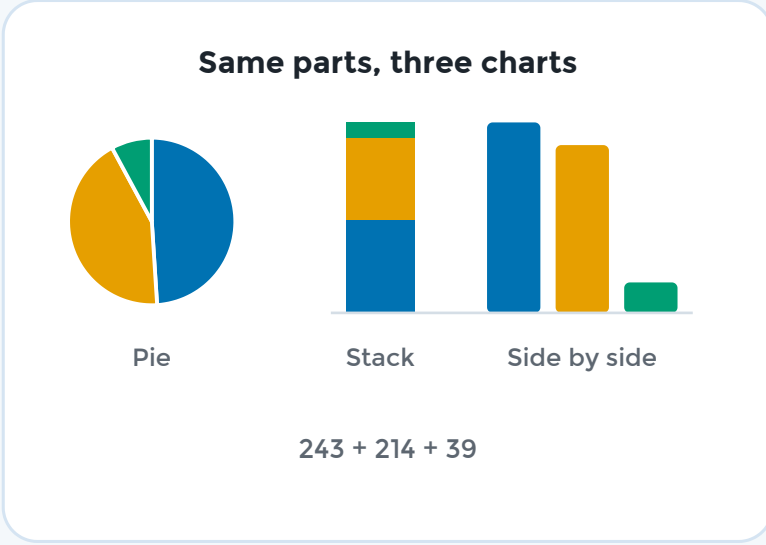
DATA 110 Week 10

Visualizing proportions

One whole, drawn three ways

Based on Chapters 10 and 11 of *Fundamentals of Data Visualization*



Prof. Rebin Muhammad
Montgomery College

TODAY

A proportion is a part of a whole

- Divide the part by the total. 243 seats out of 496 is one proportion.
- The parts of one whole add up to the total.
- A pie, a stacked bar, and side-by-side bars can show the same parts. Each chart makes a different comparison easy.

ONE WHOLE

Three ways to draw the same seats

bundestag.csv. One row is one party, and seats is its count. The 8th German Bundestag, 1976–1980, 496 seats.

THE NUMBERS

243 + 214 + 39 = 496

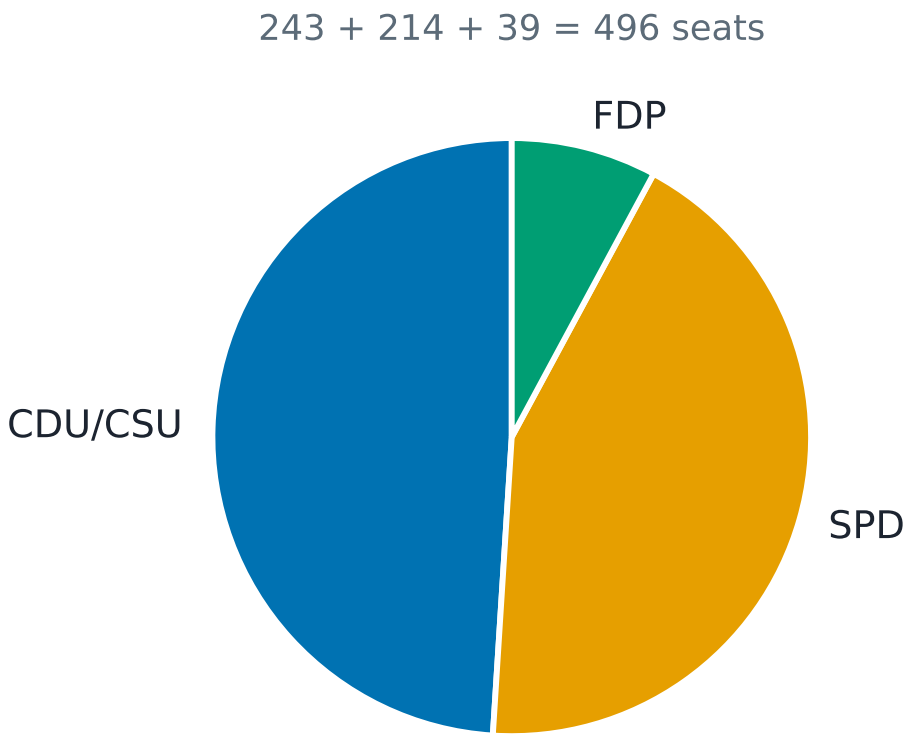
Party	Seats	Percent
CDU/CSU	243	49.0
SPD	214	43.1
FDP	39	7.9

Percent is the seats divided by 496, times 100, rounded to one decimal. CDU/CSU is just under half.

PIE

The circle is the whole parliament

CDU/CSU is the largest slice, just under half. SPD is close to it. FDP is the small slice.

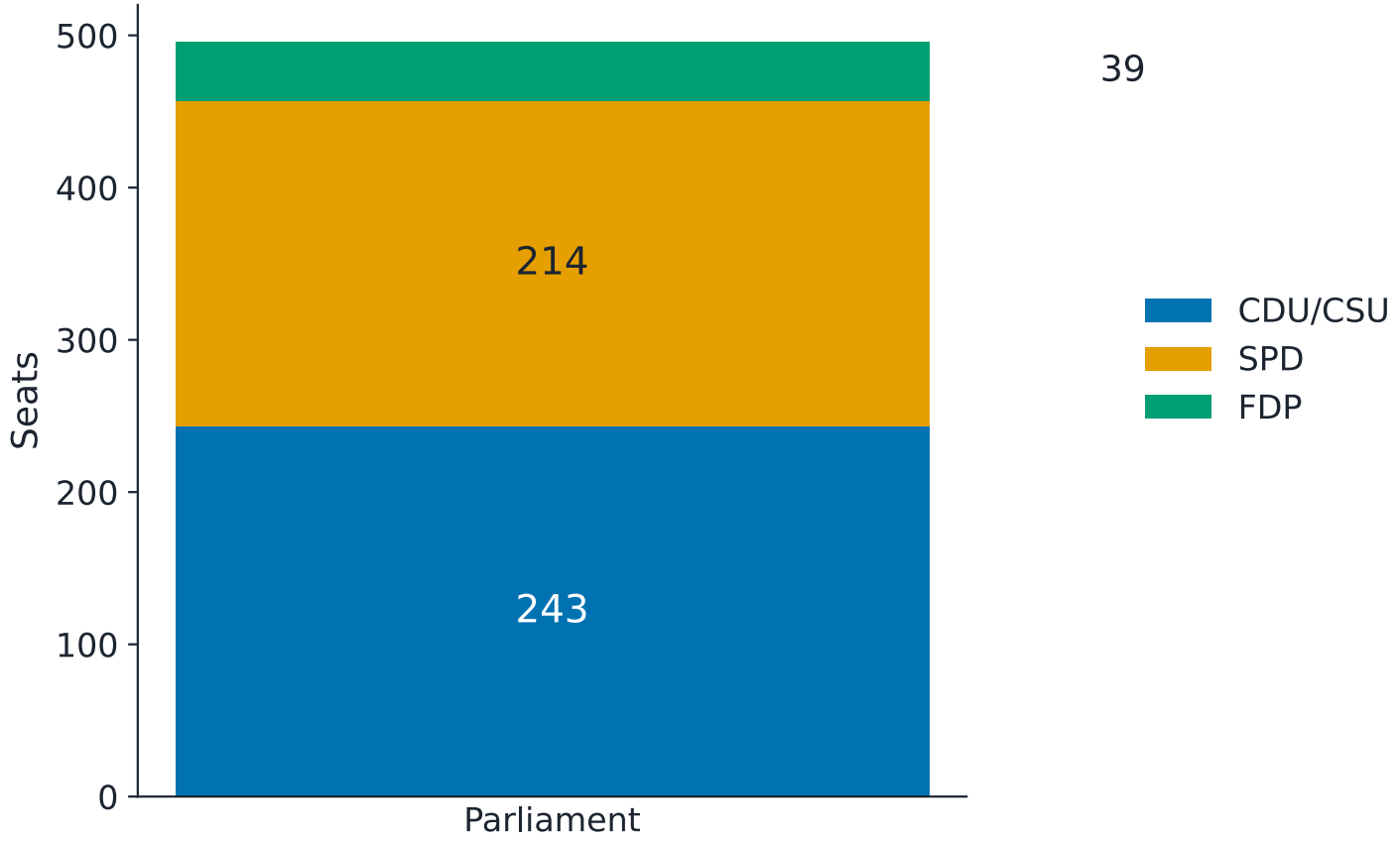


Blue is CDU/CSU, gold is SPD, and green is FDP. The same colors are used on the next two charts.

STACKED BAR

One bar is the whole parliament

The pieces sit on top of each other and add to 496.

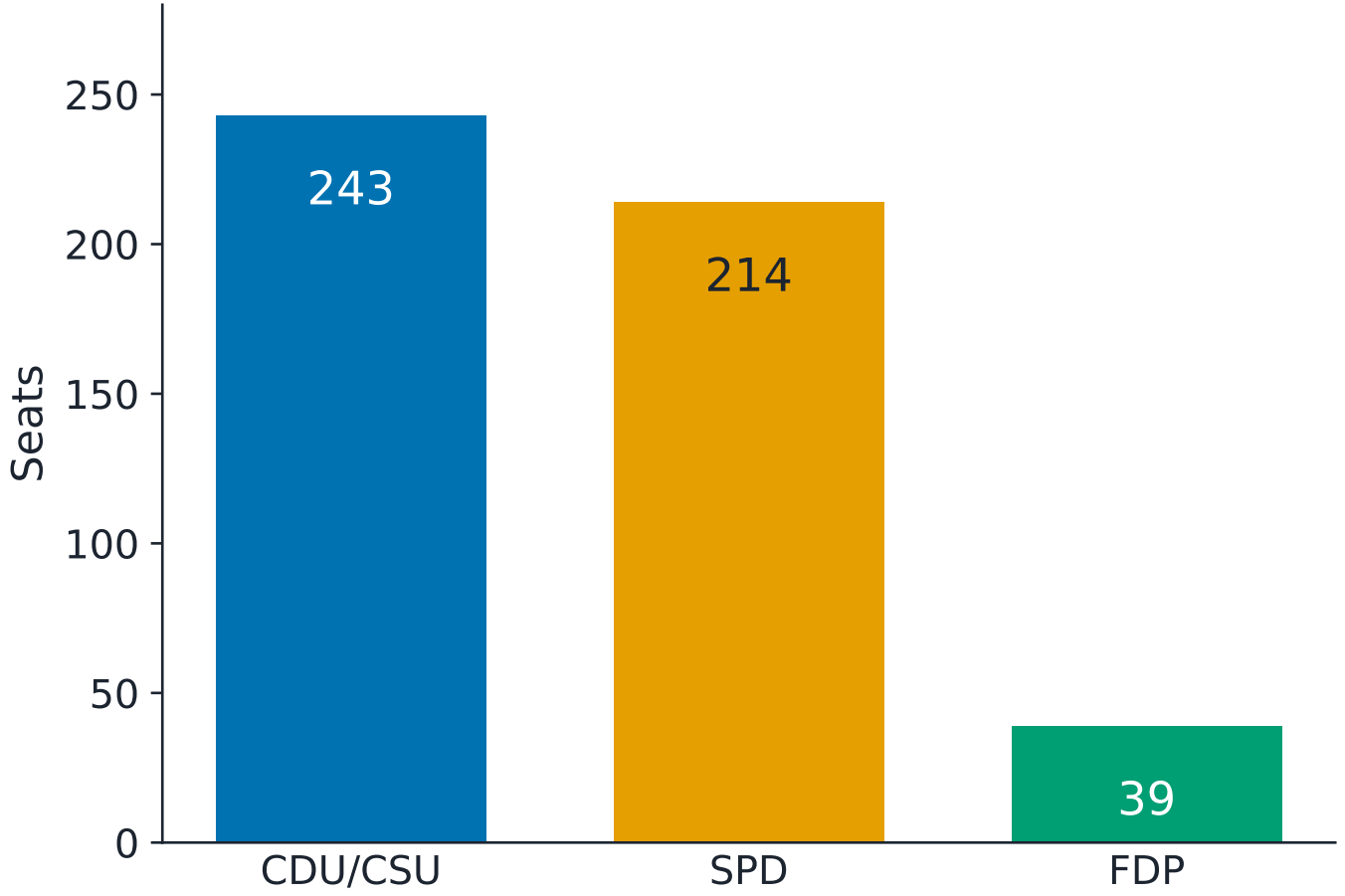


SPD starts at 243. Its piece does not share a baseline with CDU/CSU.

SIDE-BY-SIDE BARS

243 and 214 both start at zero

A shared baseline makes the two large parties easy to compare. FDP is the short bar.



QUESTION

Which chart makes 243 and 214 easiest to compare?

The same 496 seats were a pie, one stacked bar, and three separate bars.

THE COMPARISON

Side-by-side bars share a baseline

- Both bars start at zero, so the heights compare directly.
- The pie slices are close in angle. A length is easier to compare than an angle.
- On the stacked bar, SPD starts at 243.

CLOSE PARTS

Five companies, three years

marketshare.csv. Companies A through E are made up. One row is one company in one year, and each year adds to 100%.

THE NUMBERS

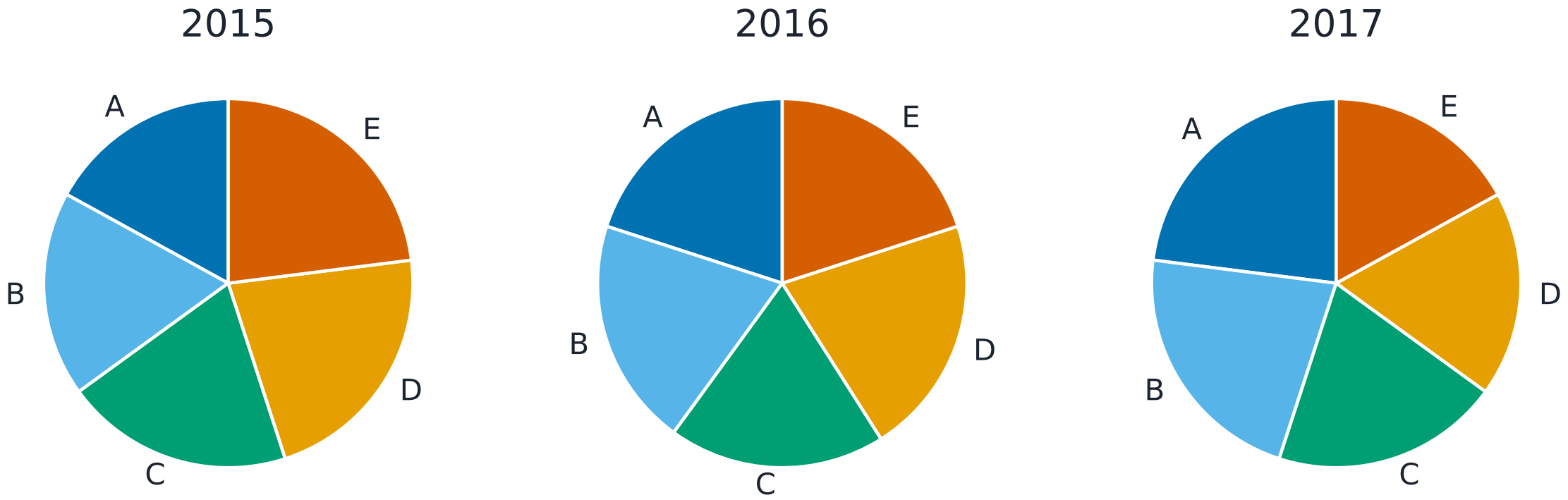
Every column adds to 100

Company	2015	2016	2017
A	17	20	23
B	18	20	22
C	20	19	20
D	22	21	18
E	23	20	17

A starts lowest and ends highest. E starts highest and ends lowest. In 2016 the five shares sit between 19 and 21.

Every slice is near one fifth

17% and 23% are hard to tell apart as angles.



Blue is A, sky blue is B, green is C, gold is D, and vermillion is E.

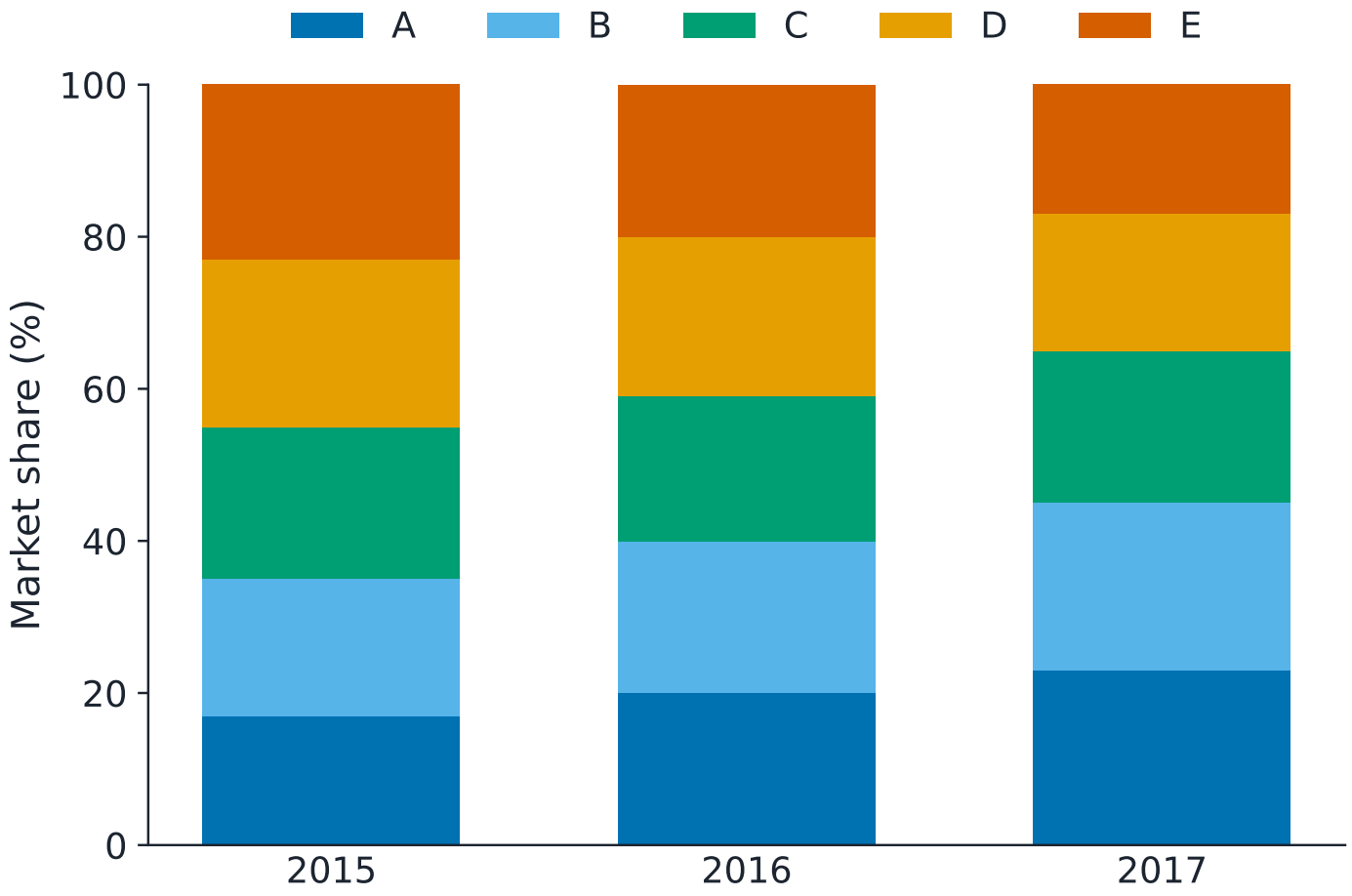
QUESTION

Which company grew from 2015 to 2017, and which shrank?

Each year is one whole market. The three pies are the three years.

The three stacks look alike

A piece in the middle of the stack does not start at zero, so the change is hard to see.



SIDE-BY-SIDE BARS

A grew, and E shrank

A goes from 17% to 23%. E goes from 23% to 17%.



Each bar starts at zero. Within one year, the five bars still add to 100%.

A CHANGING WHOLE

Women and men add to 100%

rwanda.csv. One row is one year, from 1997 to 2016. women and men are percents of the seats, and they add to 100.

QUESTION

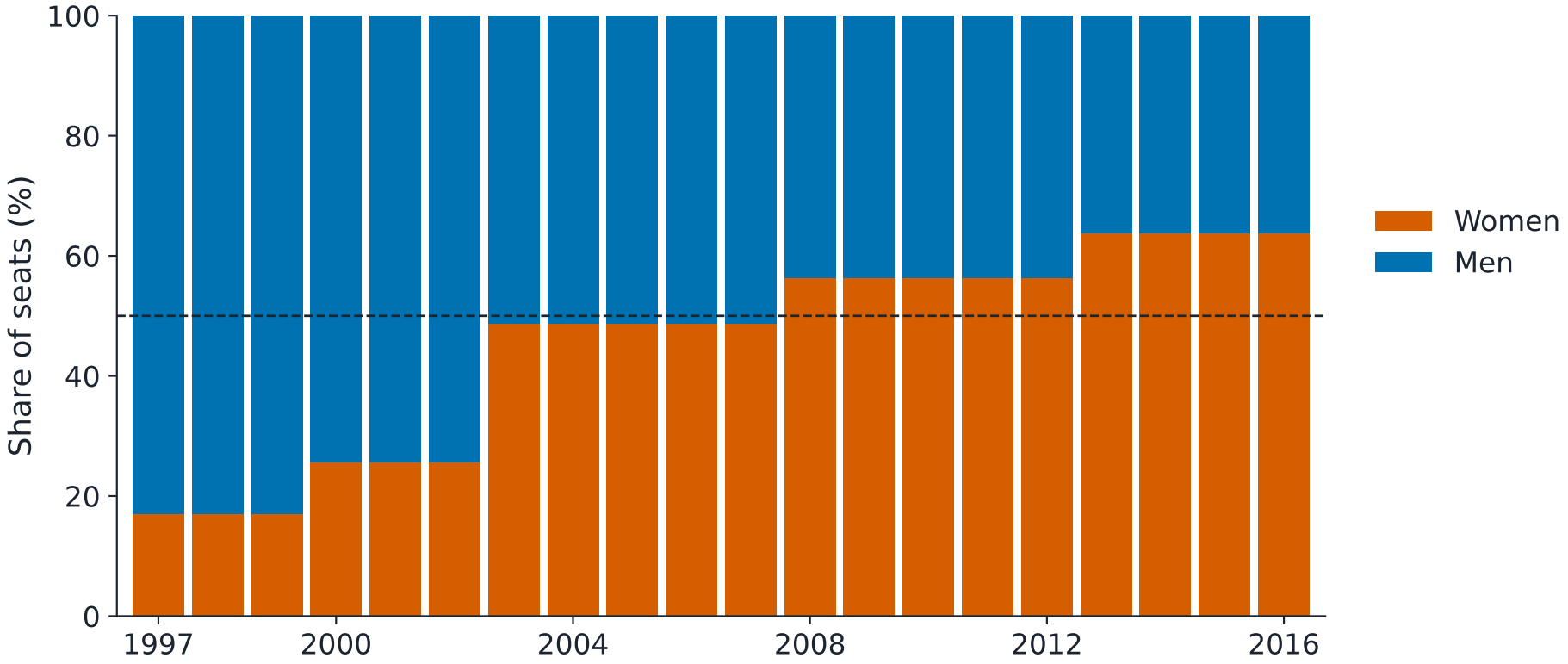
In which year do women first make up more than half?

Half is the dashed line on the next chart. Vermillion is women. Blue is men.

STACKED BARS

2008. Women reach 56.3%

In 1997, women held 17.1% of the seats. In 2003 they reached 48.8%, still under half. In 2016 they hold 63.8%.



The percentage stays flat for several years, then jumps. Each bar is one whole parliament.

CHOOSE

Each chart answers a different question

THE CHOICE

What each chart makes easy

	PIE	STACKED BAR	SIDE-BY-SIDE
Compare parts that are close in size	No	No	Yes
Show that the parts are one whole	Yes	Yes	No
Make a half, or a third, obvious	Yes	No	No
Follow one whole as it changes	No	Yes	No

The parliament comparison wanted a shared baseline. The company comparison wanted a shared baseline too. Rwanda wanted each year to stay one whole.

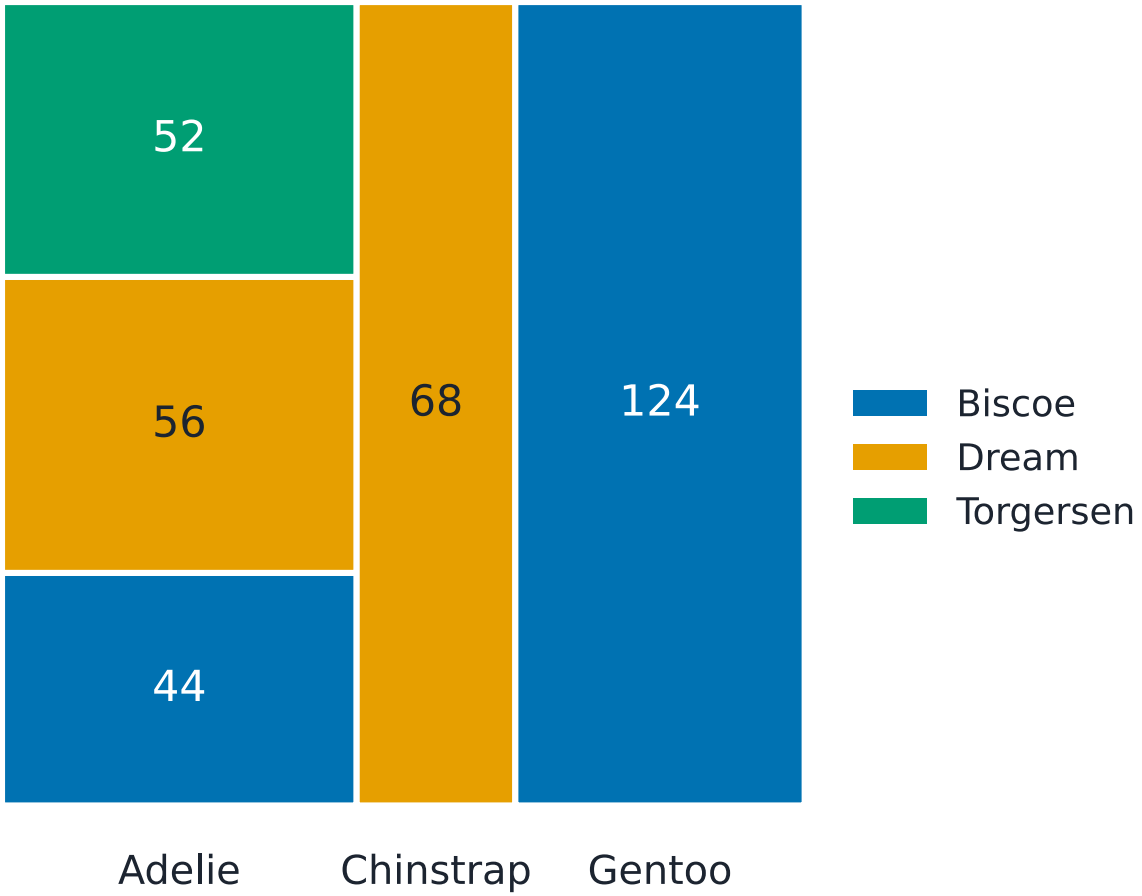
PARTS OF PARTS

A proportion can contain another proportion

penguins.csv, the same 344 penguins from earlier. The width of a column is that species' share. The islands split the column.

MOSAIC

Adelie is three islands. The others are one



Chinstrap's 68 penguins were all on Dream. Gentoo's 124 were all on Biscoe. Adelie is 44 on Biscoe, 56 on Dream, and 52 on Torgersen. Blue is Biscoe, gold is Dream, and green is Torgersen.

pie draws one slice per number

```
parties = ["CDU/CSU", "SPD", "FDP"]
seats = [243, 214, 39]
colors = ["#0072B2", "#E69F00", "#009E73"]
ax.pie(seats, labels=parties, colors=colors,
       startangle=90)
```

The circle is the sum of the seats. startangle=90 starts the first slice at the top.

bottom is where the next piece starts

```
ax.bar(rwanda["year"], rwanda["women"],
       color="#D55E00", label="Women")
ax.bar(rwanda["year"], rwanda["men"],
      bottom=rwanda["women"],
      color="#0072B2", label="Men")
ax.axhline(50, color="#1c2430", linestyle="--")
```

bottom starts the men at the top of the women bar. axhline draws the line at half.

SIDE BY SIDE

Shift each company by a fraction of a bar

```
x = np.arange(3)
width = 0.15
for i, company in enumerate(companies):
    rows = share.loc[share["company"] == company]
    rows = rows.sort_values("year")
    ax.bar(x + (i - 2) * width, rows["percent"],
           width=width, color=colors[company],
           label=company)
```

np.arange(3) is the position of each year. (i − 2) × width is −2, −1, 0, 1, 2 bar-widths, so the five companies sit beside each other.

FURTHER READING

The two chapters

- Chapter 10: Visualizing proportions
- Chapter 11: Visualizing nested proportions

Figures for Data 110, Fall 2026. Run visualizing_proportions.py to rebuild every figure.